

INDIA RUNS · DATA & AI CHALLENGE

Intelligent Candidate Discovery & Ranking

Hybrid offline ranker — semantic fit, trap-aware, production-scale

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Official bundle: candidates.jsonl (100K) · job_description.docx · redrob_signals_doc.docx · submission_spec v4
Problem: Intelligent Candidate Discovery & Ranking — top-100 shortlist for Senior AI Engineer @ Redrob AI.

Solution Overview

What we built and why it beats keyword ATS

Proposed Solution

Rank top 100 from official candidates.jsonl per submission_spec v4 (CSV: candidate_id,rank,score,reasoning).

Reads full candidate_schema: profile, career_history, skills, redrob_signals (23 fields).

Hybrid offline ranker on CPU in ~49-60s — meets 5 min / 16GB / no-network constraints.

Grounded 2-sentence reasoning per row — matches Stage 4 review criteria in submission_spec.

Differentiation vs Keyword Matching

Bundle README warns: keyword stuffers, behavioral twins, ~80 honeypots — we defend all three.

Beyond keywords: career semantics + IR skill trust + Redrob behavioral modifier.

Mirrors JD disqualifiers: research-only, consulting-only, CV/speech-only, framework-noise profiles.

open_to_work_flag=false cannot reach top-10 (per redrob_signals_doc availability logic).

Shipper-first: production-scale CPU ranker, not per-candidate LLM API calls.

JD Understanding & Candidate Signals

Official: job_description.docx + candidate_schema.json + redrob_signals_doc.docx

From job_description.docx

Role: Senior AI Engineer — Founding Team @ Redrob AI (Series A talent intelligence).

Location: Pune/Noida preferred; India Tier-1 cities; hybrid work mode.

Experience: 5–9 years band (flexible if other signals strong).

Must-have: production embeddings retrieval, vector DB/hybrid search, Python, ranking eval (NDCG/MRR/MAP).

JD rejects: pure research-only, LangChain-only AI, consulting-only careers, CV/speech/robotics without IR.

candidate_schema.json fields

profile: title, headline, summary, location, years_of_experience, company.

career_history: role descriptions with production IR evidence.

skills: proficiency, endorsements, duration_months (trust signal).

redrob_signals: 23 behavioral fields — modifier on top of skill-match score.

23 Redrob signals we weight (redrob_signals_doc.docx)

Availability: open_to_work_flag, last_active_date, notice_period_days.

Responsiveness: recruiter_response_rate, avg_response_time_hours.

Engagement: profile_views_received_30d, saved_by_recruiters_30d, search_appearance_30d.

Assessments: skill_assessment_scores (platform IR tests).

Activity: github_activity_score, interview_completion_rate, applications_submitted_30d.

Trust: verified_email, verified_phone, linkedin_connected, profile_completeness_score.

Plus: willing_to_relocate, preferred_work_mode, expected_salary_range_inr_lpa.

Ranking Methodology

Hybrid v6 scorer — retrieve, score, rank, explain

Pipeline Steps

1. Parse official job_description.docx → jd_config.py + JD embedding.
2. Load candidates.jsonl (100K per bundle README → CandidateIndex.
3. Score with hybrid v6 + redrob_signals modifier + honeypot traps.
4. Calibrate → sort → ranks 1-100; tie-break candidate_id ascending (spec §3).

Models & Algorithms

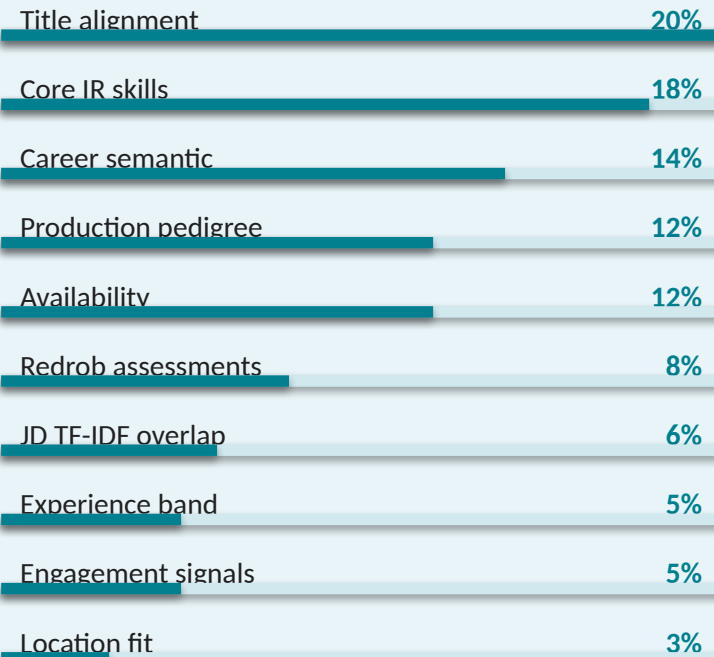
Bi-encoder: all-MiniLM-L6-v2 (committed embeddings.fp16.npz, 100K vectors).

TF-IDF JD overlap for lexical alignment with parsed JD.

Heuristic modifiers: consulting penalty, research-only cap, FAANG-current note.

Cross-encoder: OFF (RANKER_USE_CROSS_ENCODER=0) for reproducibility.

v6 Signal Weights (DEFAULT_WEIGHTS)



Explainability & Data Validation

Grounded reasoning + trap defense

Reasoning (submission_spec \$3 Stage 4)

Format matches spec example: specific facts + JD connection + honest concerns.

Cites only fields present in candidate profile — no hallucinated employers/skills.

Penalties avoided: empty reasoning, identical strings, templated name-swaps.

Rank tone matches position: strong language at rank 5, honest gaps at rank 95.

Validated: mock_stage4_review.py — 10/10 sampled rows pass 6/6 checks.

Dataset Traps (bundle README + signals doc)

~80 honeypots: impossible tenure, expert skills at 0 months (tier-0 in ground truth).

Keyword stuffers: weak titles (HR Manager, Accountant) + inflated AI skill lists.

Behavioral twins: same profile, different redrob_signals — availability wins.

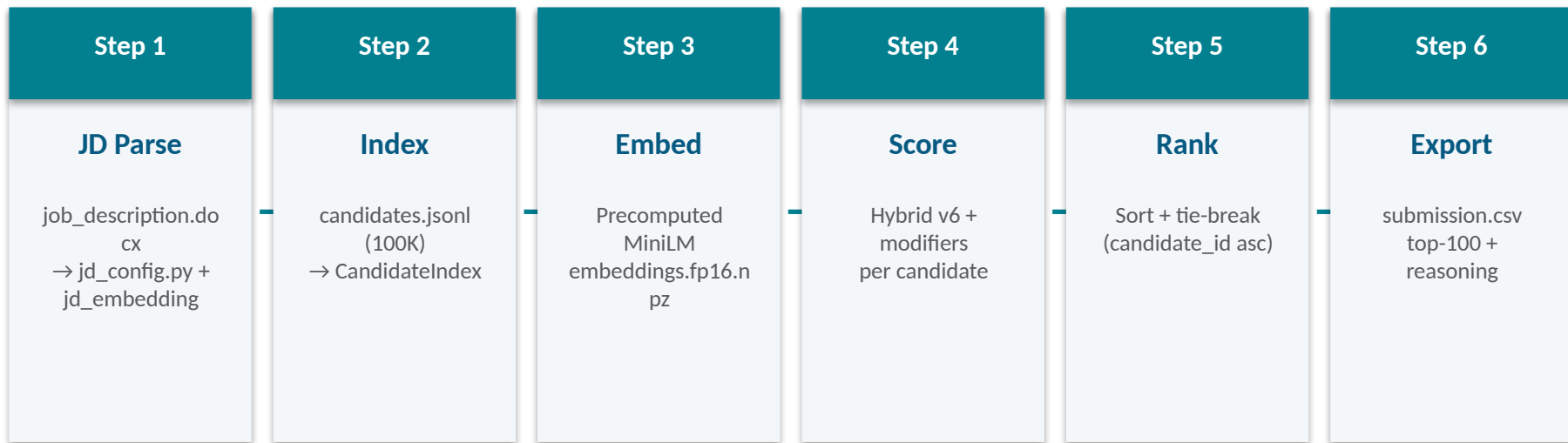
Template blurbs: recycled career descriptions across thousands of profiles.

sample_submission.csv is format-only (ranks honeypots) — not quality reference.

Our result: 0 structural honeypots in top-100 (spec disqualifies >10%).

End-to-End Workflow

JD input → ranked CSV output



Reproduce: `./scripts/reproduce_ranking.sh · python rank.py --candidates ./data/candidates.jsonl --out ./submission.csv`
Byte-verified: `scripts/verify_submission_artifact.py` confirms submission.csv matches fresh rank.py output.

System Architecture

Submission ranker (graded) vs demo platform (optional)

GRADED PATH — produces submission.csv

rank.py

redrob_ranker.py

features.py

embeddings.py

honeypot.py

availability.py

+ career_blurb.py (template dedup) · assessment.py (IR skill tests) · rerank.py (CE off)

Input: candidates.jsonl + embeddings.fp16.npz → Output: submission.csv (100 rows)

DEMO PATH — recruiter UI (not graded)

frontend/ (Next.js 14) → backend/ (FastAPI + LangGraph 7-agent pipeline) → explainability, chat, analytics

Uses top-100 import from submission.csv for dashboard — does NOT produce the graded artifact.

Results & Performance

Top picks, validation, compute compliance

Top-5 Ranked Candidates

- #1 CAND_0002025 · 0.9900 · Senior AI Engineer @ Apple
- #2 CAND_0046064 · 0.9324 · Senior NLP Engineer @ Salesforce
- #3 CAND_0010685 · 0.8659 · NLP Engineer @ Rephrase.ai
- #4 CAND_0055905 · 0.8510 · Sr ML Engineer @ Flipkart
- #5 CAND_0018499 · 0.8041 · Sr ML Engineer @ Zomato

Validation Results

100 rows · ranks 1–100 unique · scores monotonic · 0 honeypots.

All 100 candidate_ids verified in official candidates.jsonl.

Byte-reproducible artifact (verify_submission_artifact.py: PASS).

Reasoning: 100% unique, Stage 4 mock 10/10 rows pass.

~49–60s

on 100K candidates

CPU only

no GPU · no network

16 GB

RAM limit met

Well under 5-minute wall-clock limit · embeddings bundle ~68MB fp16 · Docker reproduction supported

Technologies Used

Chosen for reproducibility, scale, and spec compliance

Submission Ranker (Graded)

Python 3.11 · NumPy · PyYAML — minimal deps, no torch at rank time.

all-MiniLM-L6-v2 bi-encoder — precomputed, committed as embeddings.fp16.npz.

TF-IDF + lexical matchers for JD overlap and skill phrase detection.

Docker: docker-compose.ranker.yml for Stage 3 sandbox reproduction.

pytest: challenge/test_ranker.py (31 tests passing).

AI dev tools: Cursor + Grok (architecture/review only — no candidate data to LLMs).

Demo Platform (Optional)

Next.js 14 + Tailwind — recruiter dashboard, analytics, candidate actions.

FastAPI + LangGraph — 7-agent pipeline for interactive ranking explanations.

PostgreSQL / SQLite — workspace data; top-100 import from submission.csv.

HuggingFace Spaces — sandbox for sample ranking (≤ 100 candidates).

Vercel + Render — live product demo at recruitgpt-x.vercel.app.

Submission Assets

Everything judges need to reproduce and review

GitHub Repository

github.com/rahulx2001/recruitgpt-x

rank.py + challenge/ + validate_submission.py from bundle

Ranked CSV

[submission.csv](#) (top-100)

submission_spec v4: UTF-8, 100 rows, monotonic scores

HF Sandbox

huggingface.co/spaces/rahulsinghx2001/recruitgpt-ranker

\$10.5 — sample_candidates.json end-to-end

Official data

[candidates.jsonl](#) (100K) + [job_description.docx](#)

Synced via [scripts/sync_challenge_data.sh](#)

Reproduce

[./scripts/reproduce_ranking.sh](#)

CPU ≤5min · 16GB · no network · byte-verified artifact

Metadata

[submission_metadata.yaml](#)

Team Schadn · Rahul Kumar Singh · AI tools declared

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Thank You

Schadn · Rahul Kumar Singh

RecruitGPT X — Ranking candidates like a great recruiter, not a keyword filter.

github.com/rahulx2001/recruitgpt-x